

Abstract

Need for Scan-free State Space Models

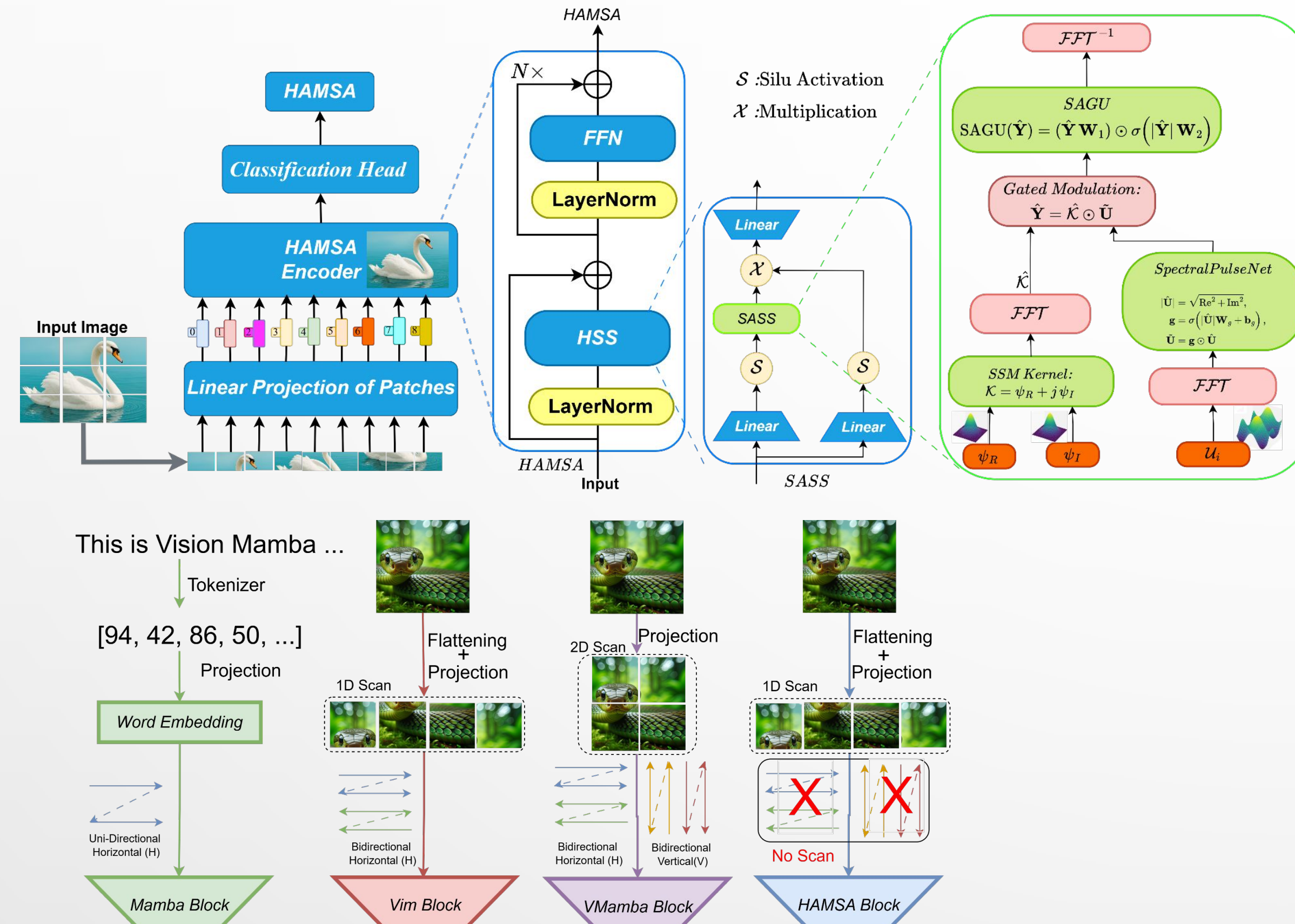
Theoretically, scanning imposes (1) **spurious causality**—assuming temporal relationships where none exist; (2) **quadratic scanning cost**—multiple passes for different directions ($\mathcal{O}(4L^2)$ for cross-scan); (3) **information bottleneck**—sequential processing limits parallelization. More fundamentally, scanning is *unnecessary*: SSM outputs are convolutions ($y=K*u$), computable via element-wise multiplication in frequency domain without any ordering. This raises our key question: *Can we directly operate in the frequency domain to eliminate scanning entirely?*

Our Insight: Scanning-Free Spectral Processing. We propose a fundamentally different paradigm: **eliminate scanning by operating in the spectral domain**. By the Convolution Theorem, SSM kernels can be computed as spectral multiplication: $y=\mathcal{F}^{-1}(\mathcal{F}(u)\odot\mathcal{F}(K))$, achieving $\mathcal{O}(L\log L)$ complexity via FFT. This spectral approach naturally suits vision: images have strong frequency-domain structure, and global dependencies emerge without sequential ordering.

Why Frequency Domain for Vision SSMs? Vision has inherent frequency structure [52]: low frequencies encode global shape, high frequencies capture fine details. Unlike language (inherently sequential), images are naturally represented in frequency domain via 2D Fourier basis. Three key advantages emerge: (1) **Global mixing**: All spatial locations interact simultaneously through frequency domain multiplication, eliminating sequential bottlenecks. (2) **Computational efficiency**: FFT enables $\mathcal{O}(L\log L)$ complexity with highly optimized GPU implementations (cuFFT), faster than $\mathcal{O}(L^2)$ attention or scanning overhead. (3) **Natural representation**: Frequency basis aligns with visual perception (contrast sensitivity functions operate in the frequency domain), enabling better inductive biases.

Theoretical Justification. Why is scanning unnecessary? SSMs model Linear Time-Invariant systems where output is a convolution: $y=K*u$. By the Convolution Theorem, this equals element-wise multiplication in the Fourier domain, where information is mixed *globally and simultaneously*—making scanning redundant.

HAMSA Model



- **First Scanning-Free Vision SSM**: Operates entirely in the spectral domain, eliminating directional scanning strategies and their computational overhead
- **Simplified Kernel Parameterization**: Direct learning of complex-valued kernel $K = \psi_{re} + j \cdot \psi_{im}$ instead of (A,B,C) matrices, removing discretization instabilities
- **SpectralPulseNet**: Novel adaptive frequency intelligence mechanism for input-dependent spectral modulation
- We show that HAMSA gets to state-of-art performance on image diverse datasets and tasks, such as image classification (ImageNet), transfer learning (CIFAR), instance segmentation (CoCo).
- HAMSA is also significantly better than other transformers and SSMs in terms of energy, memory, compute and even latency.

Contribution of HAMSA

Model	Image Size	Param (M)	MAC (G)	Acc (%)
SSMs				
TNN-T[48]	224 ²	6	-	72.3
HGRN-T[49]	224 ²	6	-	74.4
Vim-Ti[80]	224 ²	7	-	76.1
EffVMamba-T [46]	224 ²	6	0.8	76.5
PlainMamba-L1 [72]	224 ²	7	3.0	77.9
EffVMamba-S [46]	224 ²	11	1.3	78.7
PlainMamba-L2 [72]	224 ²	25	8.1	81.6
Mamba-2D-S [27]	224 ²	24	-	81.7
SiMBA-S [43]	224 ²	15	2.4	81.7
VMamba-T[34]	224 ²	22	5.6	82.2
ViM2-T [1]	224 ²	20	-	82.7
LocalVMamba-T [22]	224 ²	26	5.7	82.7
Hamsa-S(Ours)	224 ²	28	4.9	83.0
Hamsa-S* (Ours)	224 ²	28	5.0	84.1
Vim-S[80]	224 ²	26	-	80.5
EffVMamba-B [46]	224 ²	33	4.0	81.8
PlainMamba-L3 [72]	224 ²	50	14.4	82.3
VMamba-S[34]	224 ²	44	11.2	83.5
SiMBA-B [43]	224 ²	23	4.2	83.5
ViM2-S [1]	224 ²	43	-	83.7
LocalVMamba-S [22]	224 ²	50	11.4	83.7
VideoMamba-S[26]	224 ²	26	4.3	81.2
MambaTreeV-S [71]	224 ²	51	8.5	84.2
Hamsa-B(Ours)	224 ²	43	7.7	83.5
Hamsa-B* (Ours)	224 ²	43	7.8	84.9
HyenaViT-B [47]	224 ²	88	-	78.5
S4ND-ViT-B [39]	224 ²	89	-	80.4
VRWKV-B[7]	224 ²	93	18.2	82.0
VideoMamba-M[26]	224 ²	74	12.7	82.8
Mamba-2D-B [27]	224 ²	92	-	83.0
VMamba-B[34]	224 ²	89	15.4	83.9
ViM2-B [1]	224 ²	74	-	83.9
StableMamba-B [59]	224 ²	101	17.1	84.1
MambaVision-B[17]	224 ²	97	15.0	84.2
SiMBA-L [43]	224 ²	37	7.6	84.4
GroupMamba-B [55]	224 ²	57	14	84.5
MambaVision-L2[17]	224 ²	241	37.5	85.3
MambaTreeV-B [71]	224 ²	91	15.1	84.8
Hamsa-L (Ours)	224 ²	72	14.7	84.7
Hamsa-L* (Ours)	224 ²	72	14.9	85.7

Discussion

Model	Type	Params(M)	FLOPs (G)	Latency (ms)	Throughput (img/s)	Memory(GB)	Energy(J)
Convolutional Networks							
ResNet50	CNN	25.5	4.1	9.0	850	3.5	24.5
Transformer-based Models							
DeiT-S	Transformer	22.0	4.6	9.2	650	3.8	31.2
DeiT-B	Transformer	86.0	17.6	14.5	420	6.2	48.5
PVT-S	Transformer	24.5	3.8	23.8	580	4.1	35.8
Swin-T	Transformer	29.0	4.5	8.5	680	4.2	28.5
Swin-S	Transformer	50.0	8.7	10.2	590	5.8	35.8
Swin-B	Transformer	88.0	15.4	15.8	450	8.5	52.5
MLP and Pooling-based Models							
PoolFormer	Pool	31.0	5.2	41.2	420	5.5	45.8
ResMLP-S	MLP	30.0	6.0	17.4	750	4.8	28.5
EfficientFormer	MetaBlock	31.3	3.9	13.9	820	3.9	26.8
State Space Models (SSMs)							
Vim-S	SSM	26.0	8.1	7.2	780	4.5	24.8
VMamba-T	SSM	22.0	4.1	5.8	980	3.2	18.2
VMamba-S	SSM	44.0	8.7	8.9	720	4.9	25.5
VMamba-B	SSM	89.0	18.9	13.2	520	7.2	38.5
LocalVMamba-T	SSM	26.0	4.9	6.2	950	3.3	19.5
LocalVMamba-S	SSM	50.0	10.2	9.5	690	5.1	28.2
SiMBA-S	SSM	15.0	4.2	5.1	1180	2.5	15.8
SiMBA-B	SSM	23.0	8.5	7.5	890	3.8	22.1
SiMBA-L	SSM	37.0	15.1	11.2	650	5.5	32.5
HAMSA-S (Ours)	SSM	28.0	4.5	4.2	1250	2.1	12.5
HAMSA-B (Ours)	SSM	43.0	8.9	6.8	980	3.2	18.3
HAMSA-L (Ours)	SSM	72.0	15.4	9.5	720	4.8	26.8

Table 1: **Scanning strategies in vision Mambas.**

Model	Scanning Type	Scanning Strategy			Performance @ 224 ²		
		Direction	Axes	Pattern	Param (M)	FLOPs (G)	Top-1 (%)
Vim-B [80]	1D	BD	H	Raster	98	-	81.9
Mamba-2D-B [27]	1D	BD	H/V/D	Raster	92	-	83.0
EfficientVMamba-B [46]	2D	BD	H/V	Raster	33	4.0	81.8
PlainMamba-L3 [72]	2D	BD	H/V	Zigzag	50	14.4	82.3
VMamba-B [34]	2D	BD	H/V	Raster	89	15.4	83.9
LocalVMamba-S [22]	1D	BD	H/V	Local	50	11.4	83.7
SiMBA-L [41]	1D	SD	H	Raster	37	7.6	84.4
HAMSA-L (Ours)	1D	—	—	None	72	14.7	84.7

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